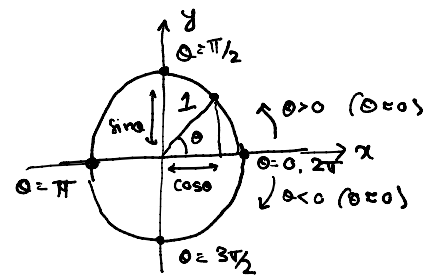


Mathematical Preliminaries

Reference: ECE 485, Duke University
 Henry Pfister (Lecture Notes)
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 "Math Review"

I(a) Trigonometric Identities

θ : angle / phase
 unit: radians
 (rad)

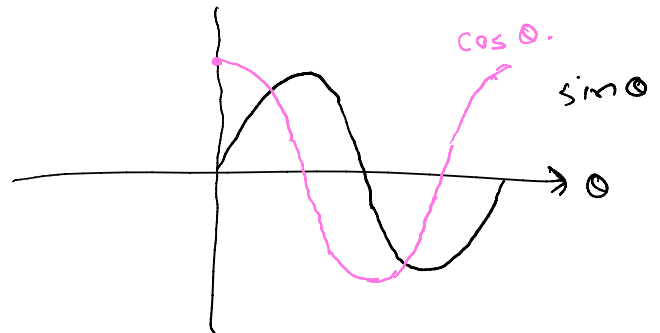


θ angle / phase

θ and $\theta + 2\pi n$ are one and the same.
 for any integer n .

(integer, $\mathbb{Z} = \{ \dots, -1, 0, 1, 2, \dots \}$)

θ	$\sin \theta$	$\cos \theta$
0	0	1
$\pi/4$	$1/\sqrt{2}$	$1/\sqrt{2}$
$\pi/2$	1	0
π	0	-1



$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi$$

$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$$

$$\sin(-\phi) = -\sin \phi$$

$$\cos(-\phi) = \cos \phi$$

$$\Rightarrow \tan(-\phi) = -\tan \phi$$

$$\tan \theta = \sin \theta / \cos \theta \Rightarrow \tan(-\phi) = -\tan \phi$$

$$\cos \theta \cos \phi = \frac{\cos(\theta - \phi) + \cos(\theta + \phi)}{2}$$

$$\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}, \quad \cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$$

I (b) Polar Coordinates.

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned}$$
$$\gamma \geq 0, \quad 0 \leq \theta < 2\pi$$

Angle: $\sin(0 + \pi) = -\sin 0$

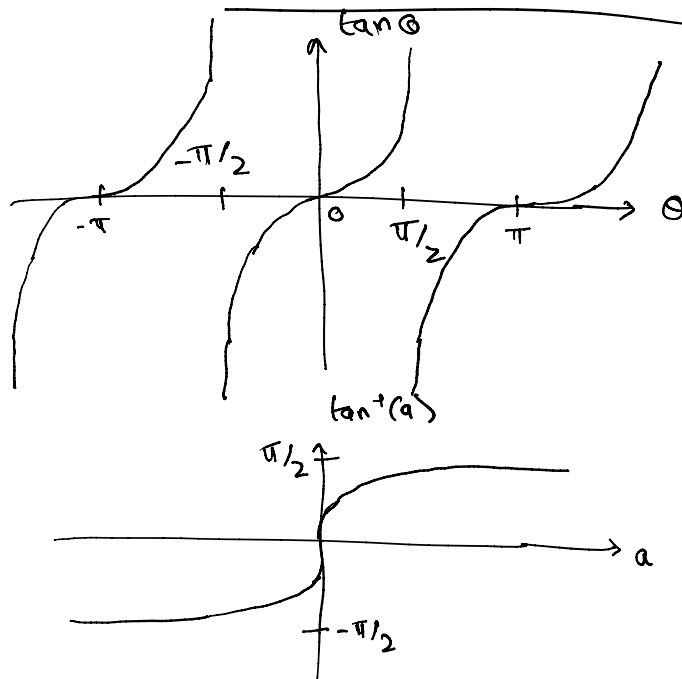
$$\cos(\theta + \pi) = -\cos\theta$$

$$\tan(\theta + \pi) = \tan\theta$$

Given (x, y) :

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}(y/x) ??$$



$$r = \sqrt{x^2 + y^2}$$

$$\theta = \begin{cases} \tan^{-1}(y/x), & x > 0 \\ \pi + \tan^{-1}(y/x), & x < 0 \\ \pi/2, & x = 0, y > 0 \\ -\pi/2, & x = 0, y < 0 \end{cases}$$

For $x=0, y=0 \Rightarrow r=0, \theta \rightarrow$ no restriction

II. Complex Numbers

We will use j to denote the imaginary number $\sqrt{-1}$.

$$x, y \in \mathbb{R}$$

$$\sqrt{-1}.$$

A complex number $z = x + jy$, $x, y \in \mathbb{R}$.

$\mathbb{C}$: Set of all complex numbers

$\mathbb{R}$: " " " " real numbers.

$\mathbb{R} \subset \mathbb{C}$ (every real number is also complex)

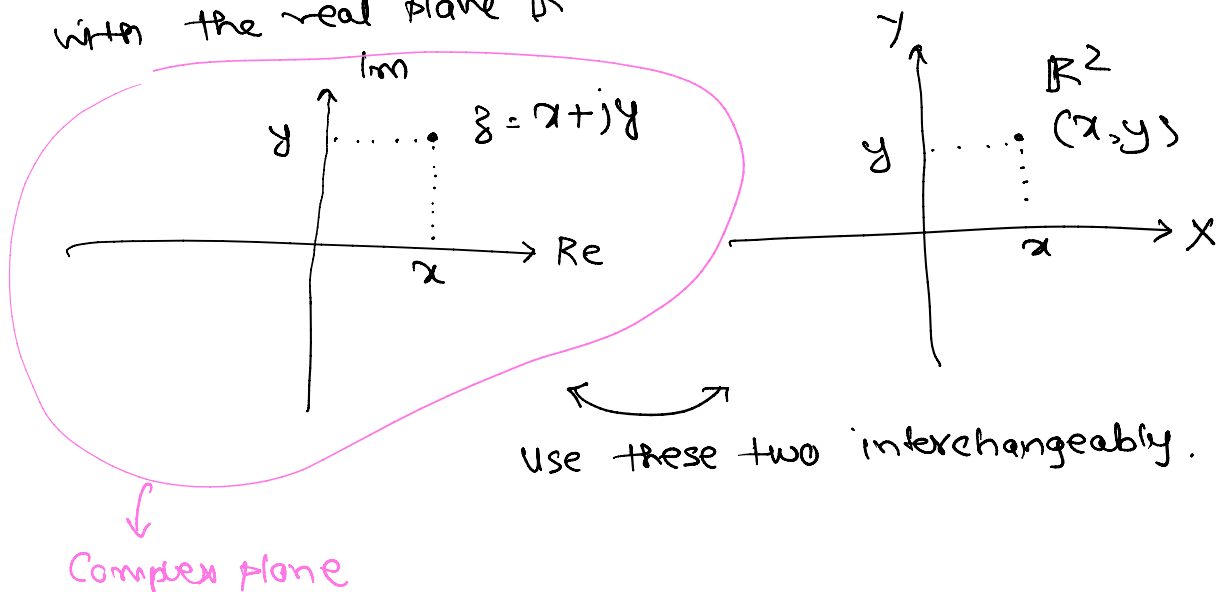
$$\hookrightarrow x = x + j0.$$

$$z = x + jy$$

x : real part of $z = \operatorname{Re}(z)$

y : imaginary part of $z = \operatorname{Im}(z)$.

The set of all complex numbers is often identified with the real plane $\mathbb{R}^2$



(a) Polar and Cartesian Forms of Complex numbers.

$$\begin{aligned} z &= x + jy \\ &= x \cos \theta + jy \sin \theta \end{aligned} \quad \left. \begin{aligned} (x, y) \\ (x \cos \theta, y \sin \theta) \end{aligned} \right\} \text{ Polar form}$$

	Name	Notation
x	Real part	$\text{Re}(z)$
y	Imaginary part	$\text{Im}(z)$
r	magnitude	$ z $
θ	Phase or angle	$\angle z, \angle z, \arg(z)$

$x+jy \rightarrow$ Cartesian form

$re^{j\theta} \rightarrow$ Polar form

Examples.

Cartesian	Polar
1	$e^{j0} = e^{j\pi} = e^{-j\pi}$
j	$e^{j\pi/2}$
-1	$e^{j\pi}$
$-j$	$e^{j3\pi/2}$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$e^{-j\theta} = \cos\theta - j\sin\theta$$

$$\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$$

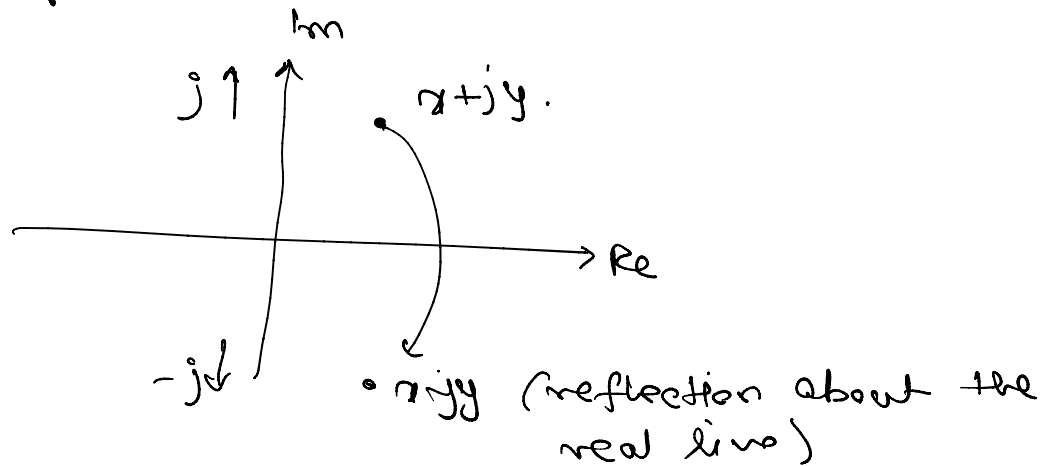
$$\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$$

(b) Conjugates. $z = x+jy = re^{j\theta}$

$$z^* = x-jy = re^{-j\theta}$$

(Replace j with $-j$ everywhere).

(Replace j with $-j$ everywhere).



(c) Arithmetic

$$z = x + jy = r e^{j\theta}$$

$$z_1 = x_1 + jy_1 = r_1 e^{j\theta_1}$$

$$z_2 = x_2 + jy_2 = r_2 e^{j\theta_2}$$

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

$$\begin{aligned} z_1 z_2 &= (x_1 x_2 - y_1 y_2) + j(x_1 y_2 + x_2 y_1) \\ &= r_1 r_2 e^{j\theta_1} \cdot e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)} \end{aligned}$$

$$\begin{aligned} (z \neq 0) \quad \frac{1}{z} &= (r e^{j\theta})^{-1} = \frac{1}{r} \cdot \frac{1}{e^{j\theta}} = \frac{1}{r} e^{-j\theta} \\ &= \frac{r e^{-j\theta}}{r^2} = \frac{x - jy}{x^2 + y^2} \end{aligned}$$

$$(z_1 + z_2)^* = z_1^* + z_2^*$$

$$(z_1 z_2)^* = z_1^* \cdot z_2^*$$

$$\left(\frac{1}{z}\right)^* = \frac{1}{z^*}$$

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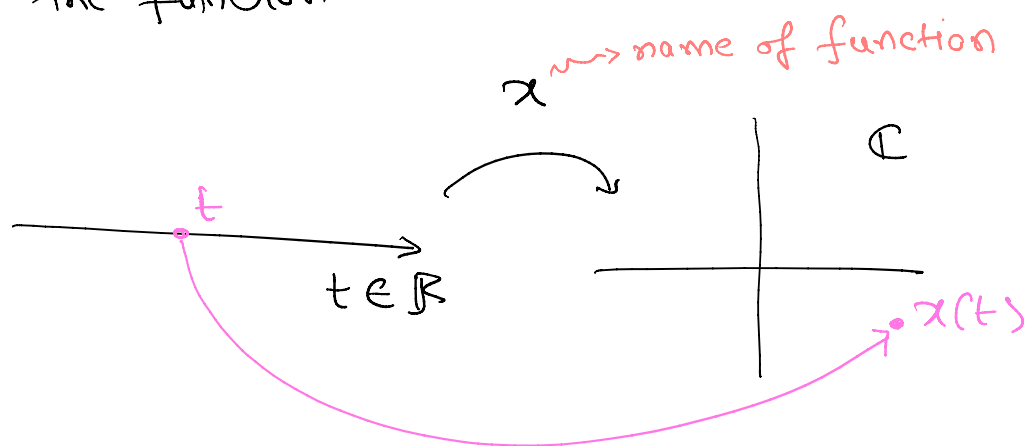
$$|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$|z_1 z_2| = |z_1| |z_2|$$

$$|1/z| = 1/|z|$$

III. Function / Signal.

In this course, we will often use t to denote the independent parameter, and the value of the function at t is, in general, complex valued.



Example: $z(t) = t^2 + j3\pi t$

$$\left. \begin{array}{l} \operatorname{Re}(z(t)) = t^2 \\ \operatorname{Im}(z(t)) = 3\pi t \end{array} \right\} \rightarrow \text{two real-valued functions of } t.$$